RETAIL SALES DATA CLEANING & WORKFLOW AUTOMATION

Project Documentation

Technology: Excel Power Query • Power Automate • OneDrive for Business • Office Scripts

1. Executive Summary

This project develops a repeatable process for preparing retail sales data for analysis using Excel Power Query and Power Automate. Power Query provides reusable data-cleaning and quality-checking logic, while Power Automate detects new files and routes supported and unsupported file types. The intended end state is an end-to-end pipeline that refreshes the transformations, validates the output, moves successful files to Processed, routes failures to Exceptions, and emails the team only after successful validation.

2. Business Problem

Retail sales files can arrive periodically and require repeatable cleaning before analysts can use them. Manual processing creates repetitive work and increases the risk of inconsistent handling of missing values, duplicate records, and transaction edge cases.

Project objectives

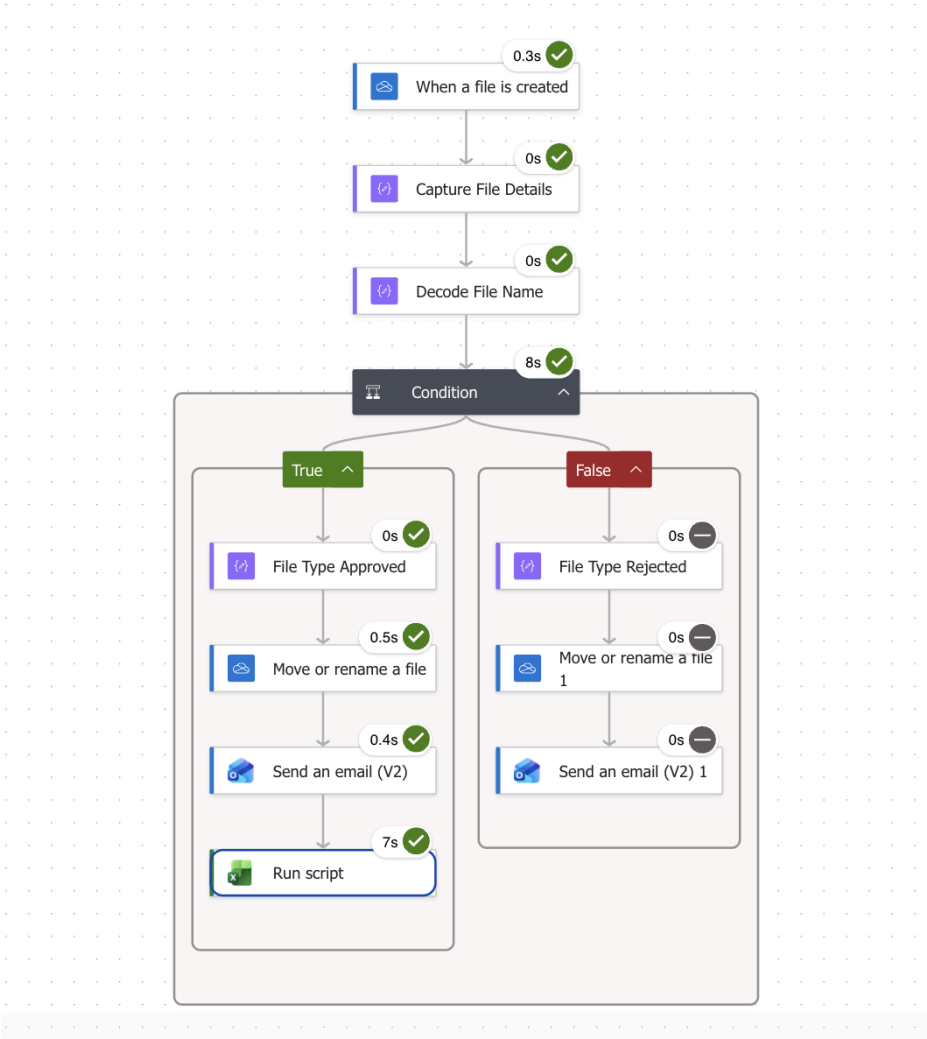
- Detects new files arriving in a designated OneDrive folder.
- Separate supported CSV/XLSX files from unsupported file types.
- Apply reusable Power Query cleaning and data-quality logic.
- Validate that the expected transformed output has been refreshed.
- Route successful and failed files to appropriate folders.
- Notify the team only when processing and output validation succeed.

3. Tools and Components

Component	Purpose
OneDrive for Business	Cloud file storage and Incoming/Processing/Processed/Exceptions folders.
Power Automate	File-created trigger, conditions, routing, orchestration, and notifications.
Excel Power Query	Data ingestion, transformation, profiling, calculated fields, and QA queries.
Master Excel workbook	Hosts the existing queries and transformed output tables.
Office Scripts	Can run supported workbook operations; the connection test alone does not refresh Power Query.

4. Target Architecture

- Incoming: A new sales file arrives.
- Power Automate trigger: Detect file creation and validate the file type.
- Processing: Stage a supported file for processing.
- Refresh mechanism: Invoke a supported, tested mechanism to refresh the workbook's Power Query transformations.
- Validation: Confirm the refresh succeeded and the expected output is current and passes required checks.
- Success: Move the file to Processed and email the team.
- Failure: Record the error, avoid a success email, and route or retain the file in Exceptions according to the retry policy.



5. Data Source and Transformation Logic

The original retail dataset includes fields such as InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

Transformation / QA check	Purpose	Handling principle
MissingDescription	Flag records with no product description.	Flag for review; do not automatically delete without a business rule.
MissingCustomerID	Identify records that cannot be attributed to a customer.	Retain when useful for sales analysis; exclude from customer-level metrics when appropriate.
Revenue	Calculate line-item revenue as Quantity × UnitPrice.	Validate numeric types and define treatment of returns/cancellations.
DataQualityStatus	Summarize relevant data-quality flags.	Make exceptions visible to downstream users.
Duplicate QA	Identify duplicate-looking rows or potential repeated events.	Define business keys and rules before removing records.
Identifier data types	Preserve invoice and stock codes correctly.	Treat codes as text when numeric conversion would damage identifiers.

Existing query names mentioned during development include src_RetailSales_Folder, stg_retailsales, fct_retailsales, and QA-related queries such as qa_duplicates / qa_retailsales.

6. Folder and Flow Design

Folder	Intended role	Rule
Incoming	New files arrive here.	Trigger starts the workflow.
Processing	Files currently being processed.	Do not treat presence here as proof of successful cleaning.
Processed	Files successfully transformed and validated.	Move here only after the success criteria pass.
Exceptions	Unsupported files or failed processing.	Keep an error reason and define whether retry is allowed.

7. Challenges Encountered and Resolutions

Challenge	What happened	Resolution / lesson
File move configuration	Destination paths did not initially work.	Use the dynamic File Identifier and the OneDrive for Business action's expected destination folder value; test the move.
Local versus cloud source	Power Query used a local OneDrive path; a cloud connection attempt returned an invalid URL/DataFormat error.	Keep query logic intact while resolving the source location for the chosen execution environment.
Run script parameter display	Power Automate showed a parameter field for a script with no intended inputs.	Save the script, recreate the Run script action, and reselect the script so its definition reloads.
Workbook not found	The Run script action could not resolve the workbook.	Use the file browser to select the actual workbook rather than relying on a stale or incorrect reference.
Script succeeded but data did not refresh	The test script only returned a confirmation string.	Distinguish connection testing from query refresh; implement and test a supported refresh mechanism.
Success status versus business success	A flow action can succeed without updated output data.	Validate the actual output tables and quality checks before moving files or notifying the team.

8. Validation and Test Plan

Test	Expected behavior	Evidence to capture
Valid CSV/XLSX arrives	Trigger fires and file enters Processing.	Flow run history and file location.
Unsupported file arrives	File is routed to Exceptions.	Flow run history and exception reason.
Missing customer/description values	Relevant flags are populated.	Sample rows in output table.
Negative quantity / returns	Values are handled according to the documented business rule.	Before/after examples and totals.
Duplicate candidates	QA identifies records according to the chosen key/rule.	QA output and documented rule.
Refresh failure	No success email; failure is recorded.	Failed run details and exception record.

Validation failure	The file is not marked Processed.	Validation result and routing evidence.
Successful end-to-end run	Output is current, file moves to Processed, and email is sent.	Run history, output evidence, and email.
Retry / repeated file	No unintended duplicate output or duplicate success notification.	Repeat-run comparison.